Elliptic Curves: Nonsingularity, Group Law, and Cryptographic Motivation

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§1.1 Definition and nonsingularity

Definition 1.1.1 (Elliptic curve in Weierstrass form)**.** An elliptic curve over a field is a nonsingular cubic curve with a chosen base point. In Weierstrass form it is written as

$$y^2 = x^3 + ax + b,$$

with discriminant nonzero.

An elliptic curve in short Weierstrass form is the set of solutions

$$E : y^2 = x^3 + ax + b$$

together with a point at infinity. The printed page emphasizes a crucial restriction: elliptic curves are not allowed to have double or triple roots in the cubic polynomial

$$x^3 + ax + b.$$

Geometrically, a triple root gives a cusp, while a double root gives a self-intersection or node. These singularities destroy the clean group law.

The nonsingularity condition is

$$4a^3 + 27b^2 \neq 0.$$

Equivalently, the discriminant

$$\Delta = -16(4a^3 + 27b^2)$$

is nonzero. The note's highlighted line is important: this condition is not cosmetic. It ensures that the tangent line is well-behaved and that the addition law can be defined without pathological exceptions.

§1.2 Why the discriminant appears

If the cubic has a double root r_1, then

$$x^3 + ax + b = (x - r_1)^2(x - r_2).$$

Expanding and comparing coefficients gives a relation between a, b, r_1, r_2. Eliminating the roots yields

$$4a^3 + 27b^2 = 0.$$

Conversely, if this expression vanishes, the cubic has a repeated root. This is the cubic discriminant calculation.

A useful conceptual memory: repeated roots occur exactly when a polynomial and its derivative have a common root. Here

$$f(x) = x^3 + ax + b, \qquad f'(x) = 3x^2 + a.$$

So singularity is the failure of f and f' to be coprime.

§1.3 Cryptographic motivation

The printed note then moves to cryptography. Earlier cryptosystems such as Diffie–Hellman use discrete exponentiation in a group. To use elliptic curves cryptographically, we need the points on an elliptic curve to form a group. The advantage is that elliptic-curve groups can give strong security with smaller parameters than classical finite-field multiplicative groups.

This is the key shift:

$$\text{number-theoretic cryptography} \quad \rightsquigarrow \quad \text{hard problems in groups.}$$

For elliptic curves, the hard problem is the elliptic-curve discrete logarithm problem.

§1.4 Geometric addition law

Let $P_1, P_2 \in E$. If $P_1 \neq P_2$, draw the line through them. It intersects the cubic in a third point R. Reflect R across the x-axis; the reflected point is defined to be

$$P_1 + P_2.$$

If $P_1 = P_2$, use the tangent line at P_1 instead of the secant line. This is point doubling.

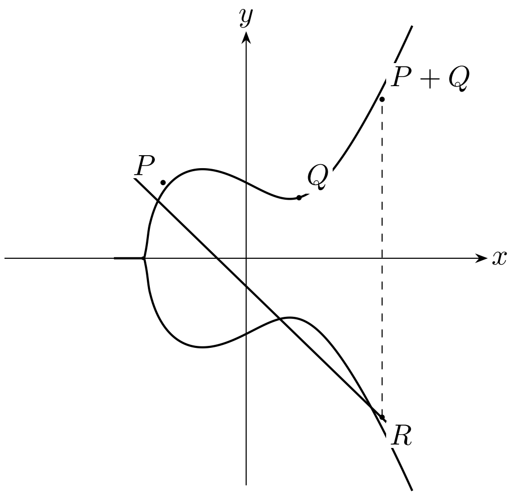

Figure 1.1: Schematic chord-and-tangent addition on an elliptic curve.

§1.5 Algebraic formula

For

$$P = (x_1, y_1), \qquad Q = (x_2, y_2),$$

with $P \neq Q$, the slope is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

Then

$$x_3 = m^2 - x_1 - x_2,$$

and

$$y_3 = m(x_1 - x_3) - y_1.$$

Thus

$$P + Q = (x_3, y_3).$$

For point doubling,

$$m = \frac{3x_1^2 + a}{2y_1}.$$

These formulas come from substituting the line equation into the cubic and using the fact that a line meets a cubic in three points counted with multiplicity.

§1.6 Special cases and the point at infinity

If $Q = -P$, then the line through P and Q is vertical, so it does not meet the affine curve in a third finite point. We define

$$P + (-P) = \mathcal{O},$$

where $\mathcal{O}$ is the point at infinity. This point acts as the identity:

$$P + \mathcal{O} = P.$$

The printed pages point out that all vertical lines on the elliptic curve meet at the point at infinity in the projective closure. This is not just a convenient fiction; projective geometry is the natural setting in which a line always meets a cubic in three points.

§1.7 Why associativity is difficult

The group law is visually simple, but associativity

$$(P + Q) + R = P + (Q + R)$$

is hard to prove by direct geometry. Algebraically it becomes a long calculation. Conceptually, elliptic curves are genus-one algebraic curves, and the chord-and-tangent law is related to divisor classes. That higher viewpoint explains why the operation is genuinely associative rather than merely plausible from the picture.

§1.8 Elliptic curves over finite fields

For cryptography one usually works over a finite field:

$$E(\mathbb{F}_p) : y^2 = x^3 + ax + b \pmod{p}.$$

The same addition formulas work, with division interpreted as multiplication by inverses modulo p. The group $E(\mathbb{F}_p)$ is finite, and scalar multiplication

$$nP = P + P + \cdots + P$$

is computationally easy, while recovering n from P and nP is believed hard for well-chosen curves.

§1.9 Why nonsingularity matters for the group law

The chord-and-tangent construction needs every line to meet the cubic in three points counted with multiplicity. If the curve has a cusp or a node, the tangent behavior at the singular point is not well-defined in the smooth sense, and the group law becomes degenerate.

Algebraically, singularity occurs when the cubic $x^3 + ax + b$ has a repeated root. This happens exactly when the polynomial and its derivative

$$3x^2 + a$$

have a common root. The discriminant condition

$$4a^3 + 27b^2 \neq 0$$

excludes that. Thus the printed discriminant is not a decorative formula; it is what guarantees that the geometric addition law behaves smoothly.

§1.10 Explicit addition formulas

If $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ with $P \neq Q$, the slope of the line through them is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

The third intersection point has x-coordinate

$$x_3 = m^2 - x_1 - x_2.$$

After reflecting across the x-axis, the sum is

$$P + Q = (x_3,\, m(x_1 - x_3) - y_1).$$

For doubling $P = Q$, use the tangent slope

$$m = \frac{3x_1^2 + a}{2y_1}.$$

These formulas are what make elliptic curves computationally usable.

§1.11 Cryptographic intuition

Over a finite field, the set $E(\mathbb{F}_p)$ is finite but still carries the elliptic-curve group law. Repeated addition gives scalar multiplication:

$$nP = P + P + \cdots + P.$$

It is easy to compute nP by double-and-add, but hard to recover n from P and nP for suitable curves and parameters. This is the elliptic curve discrete logarithm problem.

The note's cryptographic motivation should therefore be understood as a computational asymmetry: the group law is efficient in one direction and hard to invert in the other.

§1.12 Structural viewpoint

This note is a good meeting point of algebraic geometry and cryptography. A cubic equation becomes a smooth curve; smoothness is a discriminant condition; a geometric chord construction becomes a group operation; projective geometry supplies the identity element; finite fields turn the group into a computational object. The remarkable fact is that a picture of a cubic curve encodes a serious algebraic group.

§1.13 Nonsingularity and the discriminant

An elliptic curve in Weierstrass form must be nonsingular. The discriminant condition prevents cusps and self-intersections. Geometrically, nonsingularity ensures that every point has a well-defined tangent line, which is needed for the group law.

§1.14 The group law as divisor arithmetic

The chord-and-tangent rule says that three collinear intersection points add to zero. Behind the picture is divisor theory: rational functions have divisors of zeros and poles whose total degree is zero. The group law on the curve packages this linear equivalence relation.

§1.15 Associativity and geometry

Associativity is visually nontrivial because the chord construction depends on intermediate points. Algebraically it follows from the structure of the Picard group of degree-zero divisor classes:

$$E \cong \mathrm{Pic}^0(E).$$

§1.16 Cryptographic meaning

Elliptic-curve cryptography uses the difficulty of recovering n from $Q = nP$. The geometry supplies a finite abelian group over $\mathbb{F}_q$, and arithmetic security depends on the discrete logarithm problem in that group.

§1.17 Elliptic curves as geometry with a group law

Elliptic curves are not merely cubic equations. Nonsingularity gives tangent geometry, chord-and-tangent gives a group, divisor theory explains associativity, and finite-field points make the group computationally useful.